

Forecasting Hospital Bed Occupancy

A Comparative Analysis of Time Series Models

CSCI E-116: Dynamic Modeling and Forecasting in Big Data

Harvard University Extension School

December 2025

The Hospital Capacity Challenge



Understaffing Risks

- Patient safety concerns
- Staff burnout
- Quality of care issues



Overstaffing Costs

- ~25% of operating budget is nursing
- Resource inefficiency
- Budget strain

COVID-19 exposed critical gaps in capacity forecasting

AdventHealth Orlando: 197 Weeks of Data

197

weeks of data
(July 2020 – April 2024)

~2,254

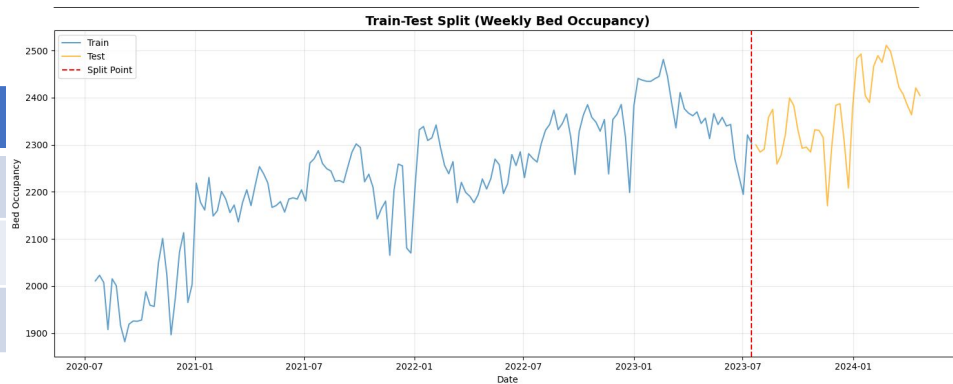
average bed
occupancy

Weekly

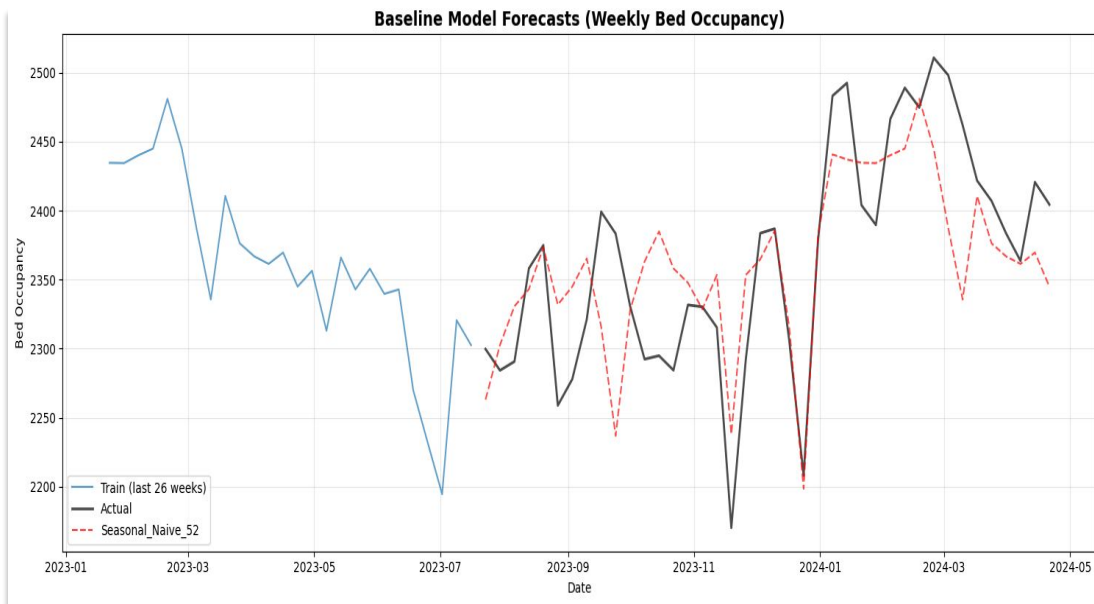
frequency

Data Sources

Source	Data
HHS	Hospital bed occupancy
CDC	Flu rates (ILI)
NOAA	Weather (temperature)



Strong 52-Week Seasonality Dominates



Trend

Pandemic dip (2020-21), then recovery to ~2,350 (2023)



Seasonal

52-week annual cycle
(winter peaks, summer troughs)



Irregular

COVID waves,
random variation



Key Insight: Annual seasonality explains most of the variation

From Simple Baseline to Deep Learning

Seasonal Naive

Baseline

Same week last year

SARIMAX

Classical

ARIMA + seasonality
+ external factors

Structural

State-Space

Explicit trend +
seasonal components

VAR

Multivariate

Joint dynamics
with COVID %

LSTM

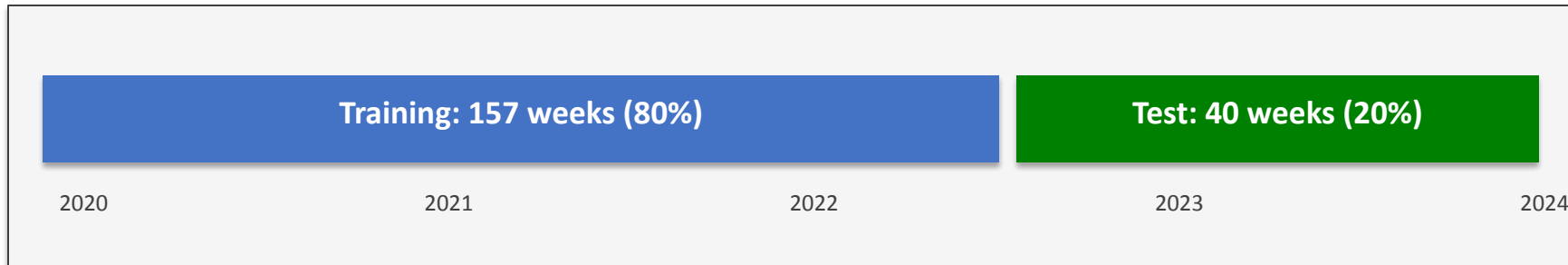
Deep Learning

Neural network
for sequences

← *Simpler*

More Complex →

Rolling Forecast with Adaptive Retraining



Key Parameters

- Train/Test Split: 80/20
- Forecast Horizon: 4 weeks
- Retraining: Every 2 weeks + error-triggered



Simulates real-world deployment conditions

Model Comparison: Test Set Performance

Model	MAE (%)	RMSE (%)	Bias (%)	Rank
Structural ★	1.75	2.21	-0.04	1
Baseline (SN52)	1.81	2.35	-0.22	2
LSTM	1.97	2.31	+0.47	3
SARIMAX	2.07	2.59	-0.40	4
VAR	2.97	3.83	+1.30	5

Errors shown as % of mean occupancy (~2,254 beds)

✓ All models achieve < 3% error = Excellent performance

What We Learned



Strong Seasonality Dominates

52-week pattern explains most variation
"Same week last year" is surprisingly accurate



Structural Model Wins

Best accuracy + near-zero bias
Directly models trend + seasonality



Complexity $\neq$ Accuracy

VAR multivariate approach underperformed
LSTM showed no clear advantage



External Factors: Modest Impact

Flu rates and weather provide marginal improvement
Internal patterns dominate

What This Means for Hospital Operations

✓ Recommended: Structural Model

- 1.75% MAE (~41 beds average error)
- Near-zero bias (well-calibrated)
- Confidence intervals for planning

Applications

Application	How It Helps
Nurse Scheduling	Plan staffing 4-8 weeks ahead
Resource Planning	Anticipate supply needs
Budget Forecasting	Project labor costs
Surge Alerts	Flag high-risk weeks



Tip: Keep Seasonal Naive as sanity check

Summary & Future Directions

Key Findings

Hospital bed occupancy is highly forecastable (~2% error)

Strong 52-week seasonality is the dominant pattern

Structural State-Space • Best accuracy (1.75% MAE) • Near-zero bias • Interpretable components

Future Directions

Explore new features (ED admissions)

Ensemble methods: Combine model forecasts for improved robustness

Real-time deployment: Clean up code. Build a production pipeline with automated data ingestion and retraining

Broader Lesson

"Model complexity doesn't guarantee better forecasts.

Understanding your data's fundamental patterns matters more than algorithmic sophistication."

References — Thank You!

Data Sources

1. U.S. Department of Health and Human Services. COVID-19 Reported Patient Impact and Hospital Capacity. healthdata.gov.
https://healthdata.gov/Hospital/COVID-19-Reported-Patient-Impact-and-Hospital-Capa/anag-cw7u/about_data
2. Centers for Disease Control and Prevention. FluView Interactive. Accessed via Delphi Epidata API. cdc.gov/fluview/overview/fluview-interactive.html
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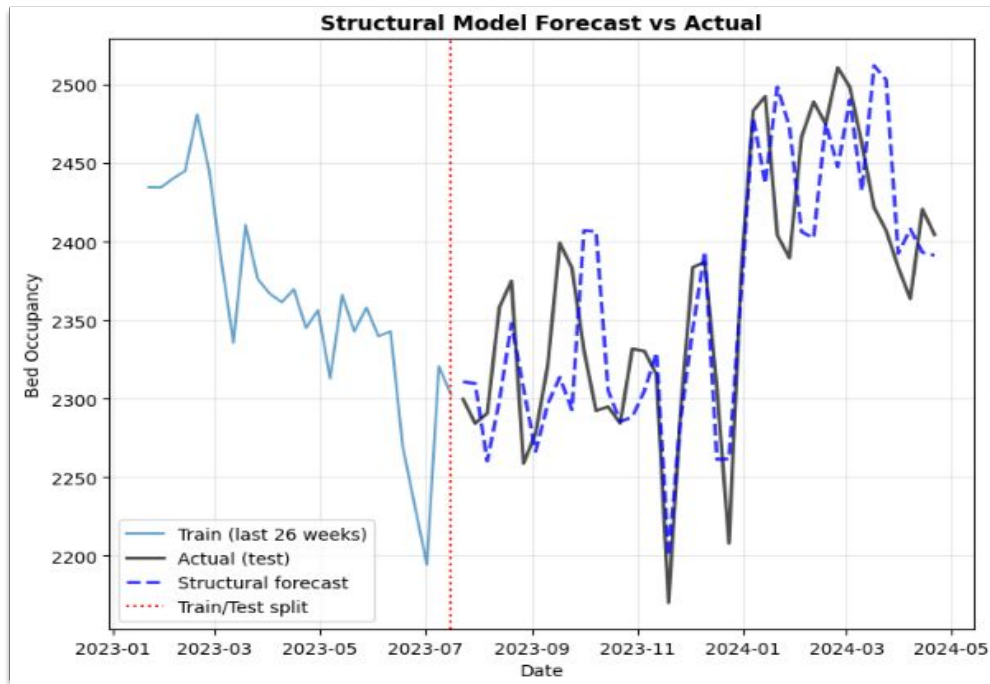
Other References

4. Hyndman, R.J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice, 3rd ed. OTexts.
5. Harvey, A.C. (1990). Forecasting, Structural Time Series Models and the Kalman Filter. Cambridge.
6. Welton JM. Hospital nursing workforce costs, wages, occupational mix, and resource utilization. J Nurs Adm. 2011 Jul-Aug;41(7-8):309-14. doi: 10.1097/NNA.0b013e3182250a2b. PMID: 21799362.

YT Link to Presentation:

<https://youtu.be/CnK2XfcPGgg>

Structural Model Wins!



Metrics

Best accuracy, Near zero bias



Seasonal

52-week annual cycle
(winter peaks, summer troughs)



Interpretable



Key Insight: Annual seasonality explains most of the variation